MSDM5004 Spring 2026

Exercise 5

Mar 16

Consider the five-point scheme for solving the following boundary value problem of the Poisson equation. Use a uniform square grid of size 1.

$$u_{xx} + u_{yy} = f(x, y), \quad 0 \leq x \leq 3,\ 0 \leq y \leq 3,$$

The boundary conditions are $u(0, y) = u(3, y) = u(x, 0) = u(x, 3) = 0$.

Write down all the individual equations of the Jacobi method for solving the linear system that results from this five-point scheme.

Solution. The five-point scheme is

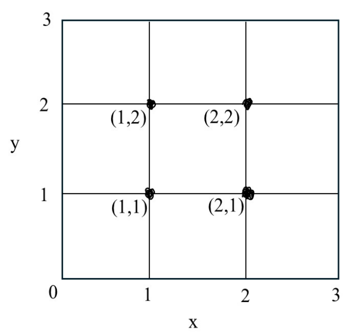
$$\frac{U_{r+1,s}+U_{r-1,s}+U_{r,s+1}+U_{r,s-1}-4U_{r,s}}{(\Delta x)^2}=f_{r,s}$$

for $1 \le r, s \le 2$.

Here $x_r = r\Delta x$, $y_s = s\Delta x$, $0 \le r, s \le 3$, with $\Delta x = 1$, and $f_{r,s} = f(x_r, y_s)$. The boundary conditions are $U_{r,s} = 0$ for $r, s = 0$ or 3.

The scheme can be written as

$$U_{r,s}=\frac{1}{4}U_{r+1,s}+\frac{1}{4}U_{r-1,s}+\frac{1}{4}U_{r,s+1}+\frac{1}{4}U_{r,s-1}-\frac{1}{4}(\Delta x)^2 f_{r,s}, \quad 1 \le r, s \le 2.$$



Using the boundary conditions, the equations are

$$\begin{aligned}
U_{1,1} &= \frac{1}{4}U_{2,1}+\frac{1}{4}U_{1,2}-\frac{1}{4}(\Delta x)^2 f_{1,1}\\
U_{2,1} &= \frac{1}{4}U_{1,1}+\frac{1}{4}U_{2,2}-\frac{1}{4}(\Delta x)^2 f_{2,1}\\
U_{1,2} &= \frac{1}{4}U_{2,2}+\frac{1}{4}U_{1,1}-\frac{1}{4}(\Delta x)^2 f_{1,2}\\
U_{2,2} &= \frac{1}{4}U_{1,2}+\frac{1}{4}U_{2,1}-\frac{1}{4}(\Delta x)^2 f_{2,2}.
\end{aligned}$$

The Jacobi scheme for solving the linear system is

$$\begin{aligned}
U_{1,1}^{n+1} &= \frac{1}{4}U_{2,1}^{n}+\frac{1}{4}U_{1,2}^{n}-\frac{1}{4}(\Delta x)^2 f_{1,1}\\
U_{2,1}^{n+1} &= \frac{1}{4}U_{1,1}^{n}+\frac{1}{4}U_{2,2}^{n}-\frac{1}{4}(\Delta x)^2 f_{2,1}\\
U_{1,2}^{n+1} &= \frac{1}{4}U_{2,2}^{n}+\frac{1}{4}U_{1,1}^{n}-\frac{1}{4}(\Delta x)^2 f_{1,2}\\
U_{2,2}^{n+1} &= \frac{1}{4}U_{1,2}^{n}+\frac{1}{4}U_{2,1}^{n}-\frac{1}{4}(\Delta x)^2 f_{2,2}.
\end{aligned}$$